Software Engineering II

Course Description

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Lecturers
Computer Engineering Program
School of Engineering
Universidad Nacional de Colombia

2026-I

Outline

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1. Course Overview
2. Syllabus
3. Bibliography

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Overview

In the **Software Engineering 2** course, students will explore *advanced development methodologies*, including the definition of functional requirements, the design of basic architectures such as `MVC` and `hexagonal`, and the implementation of `APIs` and `NoSQL databases`.

Additionally, a DevOps culture will be encouraged through `continuous integration` and `deployment practices`. Through hands-on prototypes, students will apply these concepts to develop robust and scalable software systems.

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Goals

The objective is to provide students with advanced training in the fundamentals of software development, equipping them with the *skills* needed to `design`, `implement`, and `maintain` modern software systems using agile methodologies, effective architectures, and DevOps practices.

Through *hands-on projects*, we aim to cultivate a deep understanding of modern tools and techniques in the field.

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Learning Results I

As follows, the main learning results for this course:

- **Functional Requirements**: Define, refine, and manage requirements using user stories, acceptance criteria, and story points.
- **Software Architectures**: Design and implement basic architectures such as MVC and hexagonal architecture.
- **Domain-Driven Design**: Apply DDD principles to model complex domains.
- **Design Patterns**: Apply key design patterns — *creational*, *structural*, and *behavioral* — to improve software quality.
- **Web Development**: Develop web applications covering frontend, backend, Web APIs, and HTTP communication.

Learning Results II

As follows, the main learning results for this course:

- **APIs** and **HTTP**: Design and document REST APIs following best practices.
- **Software Testing**: Implement integration testing and test-driven development (TDD) strategies using tools such as `Pytest`, `Cucumber`, and `Apache JMeter`.
- **DevOps** and **CI/CD**: Explain the DevOps philosophy and apply CI/CD practices using tools like `LocalStack`, `GitHub Actions`, `Docker`, and `Kubernetes`.

Pre-Requisites

This is a basic course, so you must have some knowledge in:

- **Programming** in Java, Python, or C++.
- **Object-Oriented Programming** concepts like classes, objects, inheritance, and polymorphism.
- **Git** basic usage, and **GitHub** basic usage.
- Use of **IDEs** like VS Code, Eclipse, or PyCharm.
- Basic knowledge of **databases** and **SQL**.

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Syllabus I

Time	Topic
Week 1	Agile Fundamentals
Week 2-3	Refining and Prototyping: *User stories, acceptance criteria (Given-When-Then), Story Mapping, Estimation, Fast Prototyping*
Week 4	Domain-Driven Design: *Hexagonal Architectures*
Week 5-6	SOLID principles: *User Stories Relation, Good Practices, Quality Attributes*
Week 7-8	Patterns Design I: *Creational*
Week 8-9	Patterns Design II: *Structural*
Week 10-11	Patterns Design III: *Behavioral*

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Syllabus II

Time	Topic
Week 11-12	Web Applications: *Frontend, Backend, Web API, HTTP communication*
Week 13	Test Engineering: *Test-Driven Design (TDD), Pytest, Cucumber, Apache JMeter*
Week 14	DevOps: *LocalStack, GitHub Actions (CI), Docker, Kubernetes, Cloud Computing*

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Grades Percentages

Item	Percentage
Quizzes	10%
Final Test	20%
Laboratories & Workshops	30%
Course Project	40%

Code of Conduct

- Always be **respectful** to your classmates and to me. You must be **kind** to everyone inside (*and outside*) the classroom.
- There is no best programming language, tool, or technology. There are only **better** or **worse** solutions.
- You must be **honest** with your work. If you don't know something, just **ask** us. We are glad to help you.
- You must be **responsible** with your work. If you don't submit **on time**, please don't complain.
- You must not be **disruptive** or **negatively affect** the **classroom environment**.

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Bibliography

Recommended bibliography:

- **Object-Oriented Analysis and Design**, by Grady Booch.
- **Design Patterns: Elements of Reusable Object-Oriented Software**, by Erich Gamma, Richard Helm, Ralph Johnson, & John Vlissides.
- **Clean Code: A Handbook of Agile Software Craftsmanship**, by Robert C. Martin.
- **The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations**, by Gene Kim, Jez Humble, Patrick Debois, & John Willis.
- **Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation**, by Jez Humble & David Farley.

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Thanks!

Questions?